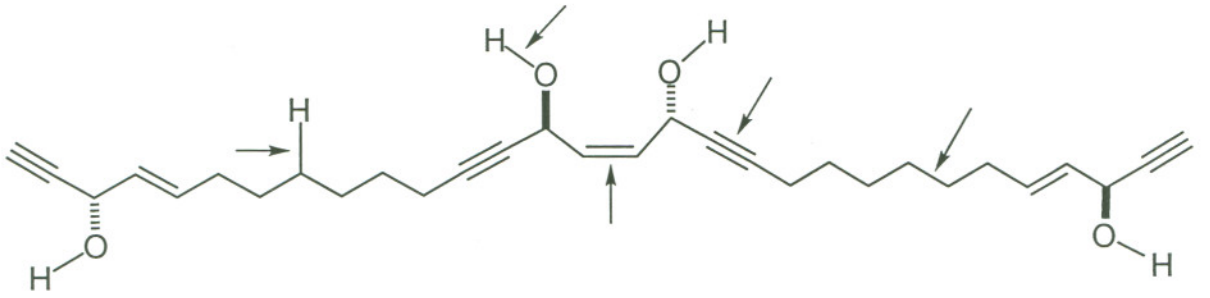
Organic Division Cumulative Exam September 13, 1997

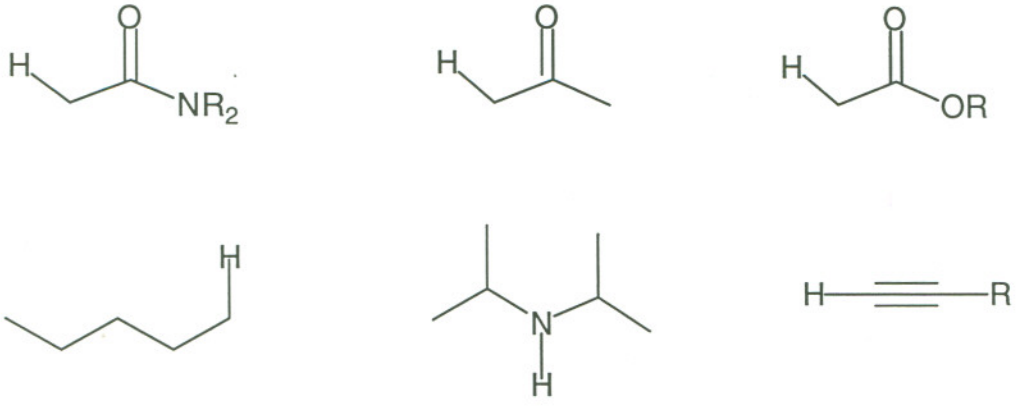
Topic: Important numbers in organic chemistry. There are not a lot of numbers you need to know in organic chemistry, but this cume will test your knowledge some of the more important ones, including bond lengths, strengths, pK_a's and A-values. If you don't know these numbers, you should familiarize yourself with them.

1. Give the bond lengths to within 0.1 Å for an average bond of the indicated type. Give the bond strengths (homolytic bond dissociation energies) rounded to the nearest 10 kcal/mol. For the multiple bonds, give the total bond energy.



A cytotoxic polyacetylene isolated from the marine sponge *Petrosia*

2. Rank the acidity from most to least acidic and give the approximate pK_a's of the following compounds.



3. Rank the substituents R according to their A-values (free energy change for the conversion of the equatorial to axial substituent) smallest to largest, and give the approximate A values (in kcal/mol).

R	A value
CH_3	
$C(CH_3)_3$	
phenyl	
OCH_3	
$\equiv$—H	

4. Give the approximate diffusion controlled rate constant of a bimolecular reaction with zero activation energy (e.g. a radical recombination reaction).